

A Skill-specific Perspective on AI-mediated Informal Digital Learning of English (AI-IDLE): Examining the Contributing Role of L2 Writing Motivation and Enjoyment



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Abstract

Artificial intelligence-mediated informal digital learning of English (AI-IDLE) is emerging as a promising field in computer-assisted language learning (CALL). This study asserts a need to adopt a skill-specific (e.g., writing) perspective to examine the unique challenges and psychological mechanisms associated with second language (L2) writing. Drawing upon the theoretical concepts of L2 writing future selves (i.e., the ideal L2 writing self and the ought-to L2 writing self; Tahmouresi & Papi, 2021) and L2 writing enjoyment (Li et al., 2023), this study seeks to investigate the motivational and affective basis of AI-IDLE-Writing. We collected questionnaire data from 412 Chinese university students and conducted post-survey interviews with 12 of them. The quantitative findings demonstrated that both the ideal L2 writing self and the ought-to L2 writing self can significantly influence AI-IDLE-Writing, but only the ideal L2 writing self can leave an impact on students' L2 writing enjoyment. Analysis of the interview data not only confirmed quantitative findings but also added nuances and contextual details to them. Furthermore, it disclosed a more dynamic and complicated mechanism of how L2 writing future selves, enjoyment, and AI-IDLE-Writing intertwine with each other as learners negotiate their own learning contexts. This explanatory sequential mixed-method design reveals implications for future pedagogical practices in supporting L2 learners' writing skill development in the environment of AI-IDLE.

Keywords: Artificial Intelligence (AI), L2 writing, writing motivation, writing enjoyment, informal digital learning of English (IDLE)

1. Introduction

The rapid advancement of artificial intelligence (AI) in recent years, particularly in the domain of generative AI powered by large language models (LLMs), has introduced unprecedented opportunities and challenges to the field of language learning and teaching (Alghamdi, 2024; Barrot, 2023a; Yang & Li, 2024). LLM-powered AI tools, exemplified by ChatGPT, can process complex linguistic patterns and generate prompt-oriented coherent responses. These tools have been found to facilitate efficient and personalised learning, provide immediate and tailored written feedback, stimulate learners' language learning motivation, and offer brainstorming-style responses (Alghamdi, 2024; Ghafouri et al., 2024; Guo & Wang, 2023; Koltovskaia et al., 2024). Such affordances have rendered language learning resources more accessible and effective for students with convenient access to AI.

In recognition of this changing reality, an increasing body of literature has explored the integration of AI technologies in second language (L2) classroom practices to enhance students' L2 learning performance (e.g., Guo & Wang, 2023; Lee et al., 2021). Moreover, a growing number of empirical studies have examined how L2 English learners engage in AI-mediated informal digital learning of English (AI-IDLE) by unpacking factors that shape L2 learners' autonomous and self-directed extracurricular learning experiences using AI (e.g., Li et al., 2024; Lin, 2024; Liu, Darvin, & Ma, 2024a, 2024b). These studies have shed light on the potential of AI in transforming language learning both within and beyond the classroom setting.

The present study is particularly interested in advancing the field of AI-IDLE because AI-IDLE offers learners a personalised and flexible learning experience, allowing them to engage with the target language at their own pace and according to their individual needs and interests. Additionally, AI-IDLE has the potential to bridge the gap between formal classroom instruction and authentic language use, providing learners with more opportunities and pedagogical support to practise their language skills in real-world contexts.

Upon a perusal of literature on AI and informal language learning (Reinders et al., 2022), there remains a gap in exploring how AI provides affordances for specific language skills and domains in the extramural and autonomous learning environment from a more nuanced perspective (e.g., L2 students use ChatGPT outside classrooms for seeking information based on their interest and receive L2 written feedback without the intervention of their teachers). We argue that L2 writing on AI-IDLE is particularly relevant and essential because writing involves a unique set of cognitive processes, such as planning, generating, and revising, which may be supported differently by AI tools compared to other language skills (Su et al., 2023). Another reason is that writing tasks often require learners to engage with complex linguistic structures and discourse features, which challenge learners' motivation and engagement in writing (Lo & Hyland, 2007). Equally important is that the increasing importance of writing development highlights the need to investigate how AI can be harnessed to help learners develop this critical skill outside of formal classroom settings (Tahmouresi & Papi, 2021; Wang & Xu, 2023). By examining the role of AI in L2 writing development through the lens of AI-IDLE, researchers can gain valuable insights into how learners engage with AI tools to improve their writing skills beyond the classroom and how AI-mediated interventions can be designed to effectively support learners' writing development. This skill-specific focus will also contribute to a more comprehensive understanding of the potential and limitations of AI in language learning and inform the development of AI-powered tools and pedagogical approaches that are tailored to the unique needs of L2 writers.

Given the above-mentioned ideas, the primary aim of this study is to investigate the motivational and affective mechanisms that shape Chinese EFL (English as a Foreign Language) learners' participation in AI-IDLE-writing. This study recognises that learners' L2

writing motivation is a crucial individual difference variable that can significantly influence their engagement and effort in L2 writing endeavours (Ghafouri et al., 2024; Lo & Hyland, 2007). In addition, L2 writing enjoyment has been identified as playing a prominent role as a positive emotional factor (Li et al., 2023; Tahmouresi & Papi, 2021; Wang & Xu, 2023), and is considered a vital affective component in language learning within and beyond the classroom (Dewaele & Dewaele, 2017; Li & Wei, 2023; MacIntyre et al., 2019). Understanding how these psychological factors interact with AI-mediated learning can provide valuable insights into sustaining L2 English learners' involvement in AI-IDLE-Writing activities. This knowledge can also contribute to the development of motivationally supportive AI tools and strategies that enhance learners' engagement with AI-IDLE practices beyond the classroom. As such, the study proposes two research questions below:

RQ1: To what extent, if any, does L2 writing future selves influence Chinese EFL learners' participation in AI-IDLE-Writing?

RQ2: To what extent, if any, does L2 writing enjoyment influence their involvement in AI-IDLE-Writing?

2. Literature review

2.1 From AI-IDLE to AI-IDLE-Writing

As a rapidly developing subfield of CALL that responds to the changing reality shaped by generative AI, AI-IDLE builds upon research on informal language learning (Lee & Lee, 2021; Lee & Lu, 2023; Zhang & Liu, 2023) and AI-assisted language instruction (Alghamdi, 2024; Barrot, 2023a; Yang & Li, 2024; Yuan & Liu, 2025). It was originally proposed by Liu and his colleagues (2024a) to examine how L2 learners acquire access to various AI tools for self-directed, independent, and autonomous language learning purposes. It refers to a broad spectrum of self-directed and extracurricular English learning activities in which L2 English learners engage by harnessing the affordances of AI technologies outside classrooms with their teachers' intervention, including generative AI platforms (e.g., ChatGPT) or ordinary learning applications that have integrated or been enhanced by AI algorithms and datasets (e.g., Duolingo). These affordances encompass features such as immediate feedback provision and automated error correction in multiple modalities (Liu & Ma, 2024; Liu, Darwin, & Ma, 2024a, 2024b). Additionally, AI-IDLE involves learners' autonomous exploration and utilisation of the extensive English learning resources associated with AI, including exposure to a wide array of English genres, styles, and registers (Guan et al., 2024a, 2024b; Liu et al., 2025). This exposure enables learners to encounter authentic language use in diverse contexts, thereby enriching their linguistic input and facilitating their L2 English development (Li et al., 2024; Liu, Darwin, & Ma, 2024b; Liu, Ng et al., 2024; Zou et al., 2025e). By participating in AI-IDLE, students can take control of their own learning, tailor their learning experiences to their individual needs and interests, and extend their language learning beyond the confines of formal classroom settings (Du & Alm, 2024; Godwin-Jones, 2024; Lin, 2024; Soyoof et al., 2023; Tram et al., 2024).

In addition to its impact on informal language learning in general, generative AI has shown remarkable potential in transforming L2 writing instruction, owing to its diverse and powerful capabilities. These capabilities include brainstorming, outline preparation, content organisation, manuscript revision, feedback provision, proofreading, and writing assessment (Guo & Wang, 2023; Shin & Lee, 2024; Steiss et al., 2024; Su et al., 2023; Zhang, 2022). A substantial body of research has corroborated that emerging AI technologies can not only facilitate learners' linguistic proficiency and writing skills (Ghafouri et al., 2024) but also boost their psychological development. For instance, AI-powered tools have been found to

increase writing confidence (Koltovskaia et al., 2024), enhance self-efficacy (Ghafouri et al., 2024), promote writing motivation and engagement (Koltovskaia et al., 2024), strengthen writing enjoyment (Barrot, 2023a), bolster autonomous writing learning (Shafiee Rad et al., 2024), reduce writing boredom (Koltovskaia et al., 2024), and alleviate writing anxiety (Alghamdi, 2024).

While previous research has primarily explored AI-mediated language writing learning in L2 classrooms (Guo & Wang, 2023; Shafiee Rad et al., 2024; Su et al., 2023; Wilson et al., 2024; Yang & Li, 2024), scholars have begun to reconsider how AI can assist L2 learners in improving their writing skills outside the classroom (Naghdipour, 2022; Zadorozhnyy & Lai, 2023). For instance, Wang et al. (2024) investigated how postsecondary L2 English learners in the United States utilised ChatGPT to aid their self-directed learning for writing purposes outside of class. The participants exhibited a strong sense of responsibility for their own learning with ChatGPT, employed various self-directed learning strategies, and reported that the advantages of using ChatGPT, such as reducing social pressure and providing immediate feedback at any time. In a similar vein, Dwina (2024) explored the potential of generative AI tools like ChatGPT in enabling Indonesian senior high school students to participate in L2 English writing practices in self-directed and out-of-class language learning environments. The findings suggested that integrating generative AI technologies could help students enhance the coherence and quality of their L2 writing. Dwina (2024) also argued that combining pedagogical approaches with AI technology has the potential to improve students' ability to communicate effectively outside the classroom.

Despite still being limited in terms of its scope, these studies have suggested that L2 English learners benefit from AI-IDLE-Writing practices as they increasingly seek opportunities to improve their writing skills outside formal educational settings. As such, understanding how they leverage AI-powered tools and resources for creative and effective out-of-class writing learning becomes crucial. By exploring learners' experiences, perceptions, and strategies in AI-IDLE-Writing, researchers can gain valuable insights into the factors that influence learners' participation in these practices and the potential benefits and challenges they face. This knowledge can also inform the design of effective AI-enhanced writing interventions and support, ultimately helping learners develop their writing skills more efficiently and productively in informal learning contexts. To offer a clear understanding and operationalization of AI-IDLE-Writing in the present study, we define it as:

AI-IDLE-Writing refers to the self-directed, out-of-class English writing activities that L2 learners engage in by harnessing AI affordances, such as instant feedback, personalised suggestions, and exposure to diverse language inputs, to enhance their writing skills, performance, and quality.

To examine the motivational and affective mechanisms that may condition AI-IDLE-Writing, we further explore the theoretical concepts of L2 writing motivation, including the ideal L2 writing self and the ought-to L2 writing self, as well as L2 writing enjoyment.

2.2 L2 writing motivation and enjoyment

Motivation, as a pivotal driving force in L2 development, significantly influences L2 writing learning (Shafiee Rad et al., 2024; Zhu et al., 2022). One of the most prominent theoretical frameworks in L2 motivation research is Dörnyei's (2009) L2 motivational self system (L2MSS), which consists of two primary components: the ideal L2 self and the ought-to L2 self. The ideal L2 self represents learners' aspirational self-images, encompassing both intrinsic integrative motivations, such as personal interest in L2 learning or the target

language community (Dörnyei, 2014; Papi, 2010; Teimouri, 2017), and extrinsic instrumental motivations, which are associated with personal goals and career aspirations (Dörnyei, 2009). Conversely, the ought-to L2 self embodies learners' sense of obligation or responsibility to engage in L2 learning, often driven by external pressures and expectations (Dörnyei, 2014). Both the ideal and ought-to L2 selves have been shown to positively impact learners' motivation and engagement in language learning activities (Teimouri, 2017). Recognising that L2 writing is a cognitively demanding task that requires a more nuanced examination of motivation (Teng & Zhang, 2018). In particular, Tahmouresi and Papi (2021) advocated a skill-specific approach to investigating L2 writing motivation. They conceptualised L2 writing motivation as learners' willingness to invest effort in improving their writing performance and their desire to achieve success in this domain. Building upon Dörnyei's (2009, 2014) L2MSS, Papi et al. (2019), and Tahmouresi and Papi (2021) emphasised the role of future L2 writing selves, namely the ideal L2 writing self and the ought-to L2 writing self, as essential sources of L2 writing motivation. Subsequent studies have confirmed the strong relationship between these future L2 writing selves and learners' writing performance (Lo & Hyland, 2007; Yu et al., 2020), as well as their emotional experiences, such as enjoyment (Zhu et al., 2022) and boredom (Solhi et al., 2024). Drawing upon Tahmouresi and Papi's (2021) conceptualisation of L2 writing selves, the current study posits that:

The ideal L2 writing self encapsulates the aspirational self-images students envision for their future as L2 learners who are able to effortlessly engage in writing tasks in the target language.

The ought-to L2 writing self represents the self-images L2 learners feel obligated to embody in their L2 writing performance due to external pressures and expectations. These self-images are characterised by the learners' perceived necessity to develop their writing skills in order to avoid negative consequences.

Enjoyment, a positive emotional factor closely associated with motivation (Lee & Lee, 2021; Tang et al., 2024; Zou et al., 2025c), refers to the pleasurable experiences individuals encounter during language learning (Li et al., 2018; Li & Wei, 2023; MacIntyre et al., 2019). Dewaele and MacIntyre (2014) define enjoyment as "a condition in which psychological needs are satisfied and forms a key element of the fundamental emotion of happiness" (p. 242). Building upon Li et al.'s (2023) conceptualization of L2 writing enjoyment, the present study proposes the following definition:

L2 writing enjoyment refers to the positive emotional experiences and feelings of pleasure that L2 learners derive from engaging in writing activities and tasks in the target language.

The enjoyable moments are transient, contributing to students' active engagement in L2 learning tasks (Liu, Darvin, & Ma, 2024b; Liu, Zhang, & Zhang, 2024, Zou et al., 2025b). Li et al. (2023) posit that L2 writing is an emotion-provoking process due to learners' relatively limited linguistic repertoire and the less automatic processes in L2 writing. However, L2 writing can be enjoyable if the writing task or practice is perceived as manageable.

2.3 Theoretical Framework

Building on L2 Motivational Self System (Dörnyei, 2009), the 2×2 Model of L2 Self-Guides framework proposed by Papi et al. (2019), was validated by extensive studies in

diverse contexts (Papi & Khajavy, 2021; Wang & Sun, 2024; Zhou & Papi, 2023). The 2×2 Model of L2 Self-Guides Framework conceptualised L2 selves into four components based on regulatory focus perspective (Papi & Khajavy, 2021): promotion self-guides (ideal L2 self and other) and prevention self-guides (ought-to L2 self and other). Among them, the ideal L2 self represents one's promotional aspirations for L2 learning achievement and pursuing desirable end-states (e.g., writing English accurately), while the ought-to L2 self represents one's prevention-oriented duties and obligations for avoiding adverse end-states (e.g., examination failure). Given the 2×2 Model of L2 Self-Guides primarily focuses on future self-images, the Proactive Language Learning Theory (PLLT) proposed by Papi and Hiver (2024) is a complementary framework that incorporates motivational and emotional predictors in shaping language learning behaviours. PLLT posits that "individuals learn a new language through proactive behaviours that maximise their engagement with the affordances of L2 interaction" (Papi & Hiver, 2024, p.9). In the context of L2 learning, proactive language learners actively engage in learning activities, thereby achieving L2 learning goals.

2.4 Research Hypotheses

While there is no study focusing on the writing motivational and writing emotional antecedents of AI-IDLE-Writing, previous research has investigated the interaction between L2 motivation and L2 enjoyment and their influence on EFL learners' participation in IDLE activities in both AI and general technology environments (e.g., Lee & Lee, 2021; Lee & Lu, 2023; Liu, Darvin, & Ma, 2024a, 2024b; Liu, Zhang, & Zhang, 2024; Zou et al., 2025a, 2025c, 2025d). The present study draws upon the findings of these related studies to formulate hypotheses regarding the relationship between L2 writing selves and emotions within the context of AI-IDLE-Writing. For instance, Lee and Lu (2023) employed a mixed-methods approach, combining questionnaire data from a sample of 417 Chinese middle-school students with interviews conducted with four participants. The findings of their study indicated that a clearly articulated ideal L2 self among L2 English learners can exert a positive influence on their L2 communication behaviours, not only within the classroom setting but also in out-of-class digital environments. Similarly, Liu, Darvin, and Ma (2024b) employed a mixed-method approach by surveying 690 Chinese undergraduates and interviewing 12 of them to examine the relationships between L2 motivation, L2 enjoyment, and AI-IDLE. Their results indicated that the ideal L2 self can positively predict both L2 enjoyment and AI-IDLE, while the ought-to L2 self only influences enjoyment. Furthermore, enjoyment was found to partially mediate the relationship between the ideal L2 self and AI-IDLE and fully mediate the impact of the ought-to L2 self on AI-IDLE. These findings have important implications for the hypotheses of the current study regarding the predictive effects of the ideal and ought-to L2 writing selves on L2 writing enjoyment and AI-IDLE-Writing practices. Specifically, the positive influence of the ideal L2 self on (AI) IDLE practices may suggest that the ideal L2 writing self may similarly contribute to learners' involvement in AI-IDLE-Writing practices. Learners with a strong vision of themselves as proficient L2 writers may be more emotionally empowered to seek out and participate in AI-mediated writing activities to develop their skills and move closer to their ideal selves. Based on these findings, we generate the following hypotheses:

H1. The ideal L2 writing self positively influences L2 writing enjoyment.

H2. The ideal L2 writing self positively influences AI-IDLE-Writing.

Although the ideal and ought-to L2 selves may exert differential effects on L2 enjoyment and AI-IDLE (Liu, Darvin, & Ma, 2024b), previous research findings still indicate

that the ought-to L2 writing self may be conducive to fostering positive emotions and sustained engagement in L2 writing practices (Tahmouresi & Papi, 2021; Wang et al., 2024). Therefore, this study still acknowledges the predictive effect of the ought-to L2 writing self on students' writing enjoyment and involvement in AI-IDLE-Writing practices. It postulates that:

H3. The ought-to L2 writing self positively influences L2 writing enjoyment.

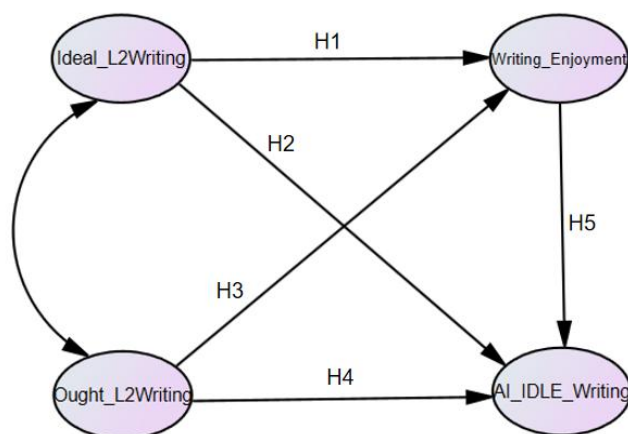
H4. The ought-to L2 writing self positively influences AI-IDLE-Writing.

Given that L2 enjoyment is considered a crucial enabler in the domain of informal language learning (Lee et al., 2021; Lee & Lee, 2021; Liu, Darvin, & Ma, 2024b; Liu, Zhang, & Zhang, 2024), this study proposes the final hypothesis:

H5. L2 Writing Enjoyment positively influences AI-IDLE-Writing.

Altogether, the relationships of these five hypotheses are illustrated visually in Figure 1.

Figure 1.
The hypothesised model



3. Methodology

This research adopted an explanatory sequential mixed methods design, consisting of a preliminary quantitative phase followed by a subsequent qualitative phase (Creswell & Creswell, 2018). The research began with the collection and analysis of quantitative data using a modified questionnaire. Drawing on the findings from the quantitative phase, semi-structured interviews were carried out to collect qualitative data to gain a deeper and more refined insight into how motivational and emotional factors influence Chinese EFL learners' participation in L2 writing activities in the out-of-class and AI-mediated environment.

3.1. Research context and participants

This study focuses on the context of Chinese EFL writing learning. Although some large language model-powered AI platforms, such as ChatGPT, are not officially accessible within China, many users managed to access them through a virtual private network (VPN) and overseas phone accounts. Moreover, several generative language model technologies have emerged in China, such as Tiangong AI, Kimi, Secret Tower Writing Cat, Literary Words from the Heart, Dragon Fruit Writing, and Deepseek.

Using convenience sampling (Creswell & Creswell, 2018), we recruited all participants online by designing an e-poster with a QR code for the online questionnaire. This poster was first distributed to our 15 colleagues who teach university English across China. Then the poster was shared with students through social media platforms such as Tencent WeChat and QQ. Although all interested learners were encouraged to open the online questionnaire, it was clearly stated that participants needed to meet the following criteria: 1) they must be registered as undergraduate, master's, or doctoral students in China; 2) they should be learning English as a foreign language; and 3) they should have prior experience using AI-assisted tools for English writing practices outside the classroom. Furthermore, participants were instructed to thoroughly review the guidelines presented in the online survey to comprehend the study's purpose.

A valid sample of 412 questionnaire responses was collected, with 142 male (34.5%) and 270 female (65.5%) respondents aged 18 to 46 years ($M = 23$). The sample consisted of undergraduate ($n = 209$, 50.7%), master's ($n = 176$, 42.7%), and doctoral students ($n = 27$, 6.6%) from the Faculties of Environment, Science, and Economy ($n = 138$, 33.5%), Humanities, Arts, and Social Science ($n = 236$, 57.3%), and Health and Life Science ($n = 38$, 9.2%). The most commonly used AI technologies were ChatGPT ($n = 283$, 68.6%), Literary Words from the Heart ($n = 188$, 45.6%), Secret Tower Writing Cat ($n = 89$, 21.6%), Cyberwriting ($n = 74$, 17.9%), Google Cloud AI ($n = 54$, 13.1%), Dragon Fruit Writing ($n = 50$, 12.1%), and Grammarly Go ($n = 34$, 8.2%). Sider had the fewest users ($n = 13$, 3.1%).

3.2 Quantitative data collection and analysis

This study implemented a revised 28-item questionnaire as the quantitative data collection tool (see Appendix A). The survey included demographic items (e.g., sex, age, educational stage, major, and locality of the university) as well as the main sections. The sources of the 28 items in the body section are detailed below.

(a) L2 writing motivation (12 items)

To measure Chinese EFL learners' ideal L2 writing self and ought-to L2 writing self, the L2 writing selves scale developed and validated by Tahmouresi and Papi (2021) was adopted. This scale, grounded in Papi et al.'s (2019) 2×2 Model, focuses on the "own standpoint" dimension, which aligns well with the objectives of this study. Papi et al.'s (2019) "own standpoint" dimensions refer to the ideal L2 self, which embodies L2 learners' own promotion-focused goals for L2 success and aspirations, and the ought-to L2 self, which represents individuals' own prevention-oriented responsibilities and obligations for evading negative outcomes. After assessing content validity in collaboration with two experienced language researchers, six items were retained for the construct of the ideal L2 writing self. Following content validity evaluation, the factor of the ought-to L2 writing self also comprised six items. The survey used a six-point Likert scale ranging from 1 (Strongly disagree) to 6 (Strongly agree).

(b) L2 writing enjoyment (6 items)

Regarding L2 writing enjoyment, we adapted Li et al.'s (2023) *Foreign Language Writing Enjoyment Scale*, which was developed and validated using a sample of Chinese EFL junior high school students ($N = 1672$). The original scale consisted of 9 items, including six items measuring the dimension of private writing enjoyment and three items examining social writing enjoyment. The dimension of private writing enjoyment refers to the satisfaction derived from writing as an individual activity, while social writing enjoyment relates to the pleasure gained through interpersonal interaction and collaborative creation (Li et al., 2023). In this study, we focused primarily on learners' private enjoyment of informal digital learning

English writing. Therefore, only the six items testing private writing enjoyment were retained to assess the writing enjoyment levels of Chinese university EFL students. The L2 writing enjoyment items were measured using a five-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree).

(c) AI-IDLE-Writing (10 items)

To implement the AI-IDLE-Writing, we primarily drew upon the scale developed by Liu and Ma (2023), which measured Chinese EFL learners' acceptance and usage of ChatGPT for IDLE purposes. The survey utilised a 6-point Likert scale, with response options ranging from 1 (Strongly disagree) to 6 (Strongly agree). Since this study specifically focuses on the L2 writing context, the original questionnaire was adapted and modified to create the AI-IDLE-Writing Scale. Following a content validity check, the revised scale consisted of 10 items.

We conducted the statistical analysis using SPSS 29.0 and AMOS 26.0, applying the Structural Equation Modeling (SEM) approach to examine the questionnaire data through five key steps (Collier, 2020; Kline, 2023). Initially, a thorough data screening and cleaning process was performed to identify outliers, missing data, and invalid responses (e.g., those completed in under 100 seconds or with an implausible age, such as 100 years). The second step involved assessing the normality of the dataset by calculating the kurtosis and skewness for each item. Third, the reliability of each construct was evaluated by calculating its Cronbach's alpha value. Subsequently, a thorough Confirmatory Factor Analysis (CFA) was performed to assess the validity of the questionnaire. Finally, based on the reliability and validity results, models for L2 writing motivation, L2 writing enjoyment, and AI-IDLE-Writing were constructed. A thorough path analysis was then conducted to explore the relationships between L2 writing motivation, writing enjoyment, and AI-IDLE-Writing.

3.3 Qualitative data collection and analysis

Following the completion of the online survey, the research team provided contact information, inviting participants to reach out to the last author if they were interested in participating in the subsequent interview. A total of twenty-four students expressed their interest by contacting the last author, but only twelve students were randomly selected to take part in the post-survey interview. In qualitative data collection, 12 semi-structured one-to-one interviews were conducted synchronously online. The purpose of these interviews was to probe the underlying reasons related to participants' motivation and enjoyment in AI-IDLE-Writing activities in L2 writing. Prior to the interviews, participants carefully reviewed and signed informed consent forms. The interview process commenced by reviewing the participants' questionnaire responses, followed by in-depth conversations guided by a semi-structured interview protocol, which was created on the basis of prior literature (e.g., Liu, Darwin, & Ma, 2023; Liu & Ma, 2023). The interview protocol was adapted to reflect the context of EFL learners' L2 writing experiences (see Appendix A). Participants were encouraged to express their thoughts and opinions in Mandarin Chinese, as the interview questions were provided in a bilingual format. The interviews were recorded and lasted between 30 to 45 minutes under the participants' permission. The discussions were designed to gain a deeper understanding of the acceptance and experiences of AI technologies in AI-IDLE-Writing.

We conducted a thematic analysis (Braun & Clarke, 2006) using open, axial, and selective coding (Corbin & Strauss, 2008) to analyze the interview transcripts. Open coding involves the initial process of categorising and labelling key concepts that emerge from a meticulous, line-by-line examination of the qualitative data, focusing on themes pertinent to

the research questions. Axial coding entails the subsequent phase of clustering and aggregating the concepts and categories identified during open coding based on their inherent properties and dimensions. The researchers engaged in a comprehensive analysis of the open coding results, identifying commonalities and distinctions, and synthesising conceptually related ideas into overarching categories. For instance, “writing interest” was classified into the category of “ideal L2 self”. Selective coding constitutes the final stage of identifying core categories and systematically establishing connections between these central themes and other related categories. For example, the researchers identified “AI-mediated informal digital L2 writing learning experience” as the core category, analysing the associations between other categories (e.g., ideal L2 writing self, ought-to L2 writing self, and writing enjoyment). The coding process involved condensing the textual data into smaller units, progressively merging codes with similar characteristics into larger categories, and extracting themes relevant to the research questions and quantitative hypotheses. To ensure the reliability and consistency of the qualitative data analysis, the following steps were taken to address inter-coder reliability. Initially, the first and last authors independently transcribed the interviews and segmented the transcripts into discrete units, assigning descriptive labels to each. Subsequently, they engaged in an iterative process of comparing and refining their coding schemes to identify and substantiate the connections between codes and to develop potential thematic categories (e.g., the role of the ideal L2 writing self in motivating students’ participation in AI-IDLE writing activities). Third, the inter-coders have regular meetings to compare their coding, discuss disagreements, and reach a consensus on the final coding decisions. Concurrently, the second and third authors acted as “unbiased third parties”, enhancing the trustworthiness and rigor of the analysis by evaluating and questioning the adequacy of the themes and their associated codes in addressing the two research questions. The Cohen’s kappa coefficient was .93, indicating a high level of reliability in the coding process. The qualitative analysis aimed to explore the underlying reasons behind the themes that emerged from the quantitative findings. In this mixed-method approach, the quantitative SEM results, based on a large sample, investigated the relationships between L2 writing motivation, L2 writing enjoyment, and AI-IDLE-Writing. The qualitative interview findings provided deeper contextualised insights into participants’ adoption of AI technologies in their AI-IDLE-Writing activities. Thus, the qualitative results not only explained the quantitative findings but also offered a more nuanced and multifaceted understanding of how Chinese EFL learners participate in AI-IDLE-Writing.

4. Findings

4.1. Quantitative results

4.1.1 Descriptive statistics

The descriptive statistics in Table 1 reported the Mean (M), Standard Deviation (SD), Kurtosis, and Skewness, factor loadings. Specifically, the M scores for the ideal L2 writing self, the ought-to L2 writing self, and AI-IDLE-Writing ranged between 3.4 and 4.8, indicating that participants had a clear ideal and the ought-to L2 writing self and relatively frequent use of AI technologies for informal writing learning activities outside the classroom. In contrast, the M scores for L2 writing enjoyment ranged between 2.9 and 3.5. The SDs for these items ranged from 0.8 to 1.4, indicating a small and acceptable level of variation in responses. Additionally, the skewness and kurtosis values for each item were displayed in Table 1, with absolute values falling within the benchmark ranges of < 3 for skewness and < 8 for kurtosis (Kline, 2023), suggesting that the dataset basically demonstrated a normal distribution.

Table 1
Descriptive statistics

Factors	Sample (N = 412)						Cronbach's a (> 0.7)
	Items	M	SD	Skewness	Kurtosis	Factor loading (In CFA)	
the ideal L2 writing self	IW1	4.8	0.9	-0.94	1.95	.60	.88
	IW2	4.4	1.0	-0.55	0.39	.72	
	IW3	4.5	1.0	-0.83	1.01	.75	
	IW4	4.6	1.0	-0.85	1.20	.76	
	IW5	4.5	1.0	-0.62	0.67	.77	
	IW6	4.7	0.9	-0.72	1.25	.83	
the ought-to L2 writing self	OW1	4.3	1.1	-0.65	0.37	.69	.81
	OW2	4.1	1.3	-0.54	-0.43	.62	
	OW3	4.4	1.1	-0.72	0.35	.62	
	OW4	4.0	1.2	-0.38	-0.27	.72	
	OW5	4.3	1.2	-0.73	0.21	.57	
	OW6	3.4	1.4	-0.05	-0.89	.65	
L2 writing enjoyment	WE1	3.3	0.8	-0.07	0.72	.63	.88
	WE2	3.0	0.8	-0.21	0.74	.82	
	WE3	3.0	0.9	-0.06	0.38	.87	
	WE4	3.7	0.8	-0.64	0.94	.61	
	WE5	2.9	0.9	-0.12	0.23	.73	
	WE6	3.5	0.9	-0.42	0.50	.76	
AI-IDLE- Writing	AI1	4.6	0.9	-1.03	2.21	.67	.91
	AI2	4.6	0.8	-0.56	0.66	.76	
	AI3	4.5	0.9	-0.96	1.73	.72	
	AI4	4.4	1.0	-0.64	0.52	.73	
	AI5	4.4	1.0	-0.75	0.64	.75	
	AI6	4.5	0.9	-0.80	1.32	.71	
	AI7	4.3	1.0	-0.74	0.44	.74	
	AI8	4.5	1.0	-1.00	1.41	.59	
	AI9	4.5	1.0	-0.96	1.59	.67	
	AI10	4.5	0.9	-0.68	0.72	.74	

4.1.2. Reliability and validity examination

Regarding the reliability examination, we used Cronbach's alpha (α) values to evaluate. As demonstrated in Table 1, the α values for the four constructs were .88 for the ideal L2 writing self, .81 for the ought-to L2 writing self, .88 for L2 writing enjoyment, and .91 for AI-IDLE-Writing, all of which exceed the recommended threshold of .7 (Kline, 2023). These results indicate that excellent reliability or internal consistency has been established for the measures used in this study.

The validity of the questionnaire was carefully inspected through tests of convergent and discriminant validity. Regarding Convergent validity, the composite reliability (CR) and the average variance extracted (AVE) for each variable were examined. Initial analysis revealed inadequate convergent validity, as the CR values for all four factors, while

surpassing the recommended cutoff of .7 (refer to Table 2), were still insufficient. The AVE value for the ought-to L2 writing self did not surpass the .5 cut-off point (Hair et al., 2009). However, after deleting items with low factor loadings (i.e., Item O3 and O5 from the ought-to L2 writing self), the recalculated AVE value for the ought-to L2 writing self approached the .5 threshold. According to Fornell and Larcker (1981) and Lam (2012), if the AVE is below .5 but close, it may still be acceptable, especially if the CR is greater than .6, which suggests that the construct's convergent validity is still sufficient. Concerning discriminant validity, Heterotrait-Monotrait Ratio of Correlations (HTMT) was tested (Henseler et al., 2015). As presented in Table 2, the HTMT values for all four factors were below .9, indicating that the variables are statistically distinct from each other.

AMOS was employed to develop a measurement model, offering further evidence to support the structural validity of the revised questionnaire. The model's fit to the dataset was evaluated using several indices, including the ratio of χ^2 to degrees of freedom (χ^2/df), Comparative Fit Index (CFI), Incremental Fit Index (IFI), Tucker-Lewis Index (TLI), Normed Fit Index (NFI), Root Mean Square Error of Approximation (RMSEA), and Standardized Root Mean Squared Residual (SRMR). Goodness-fit indices fell in the advised ranges for the measurement model, indicating that the adapted questionnaire has established construct validity (see Table 3).

Table 2

Convergent validity and discriminant validity measurement

		AVE (> .5)	CR (> .7)	HTMT (< .9)			
				1	2	3	4
1	the ideal L2 writing self	.59	.88	-			
2	the ought-to L2 writing self	.46	.77	.67	-		
3	Writing enjoyment	.64	.88	.43	.38	-	
4	AI-IDLE-Writing	.51	.88	.44	.44	.23	-

Table 3

Goodness-model fit indices of measurement model and structural model

	χ^2/df	CFI	IFI	TLI	NFI	RMSEA	SRMR
The measurement model	2.55	.94	.94	.93	.902	.06	.05
The 1 st structural model	2.99	.88	.88	.87	.83	.07	.06
The 2 nd structural model	2.53	.94	.94	.93	.903	.06	.05
Threshold values (Kline, 2023)	< 3	> .90	> .90	> .90	> .90	< .10	< .08

4.1.3. Hypothesis testing

We examined the structural model twice to test the hypotheses. The initially proposed structural model did not fit the dataset well, as four of the seven indices failed to meet the recommended benchmark values. In response, the structural model was refined by removing items with lower factor loadings across different dimensions and calculating modification indices (MI) in AMOS for further adjustments. The adjusted structural model demonstrated a

good fit with the dataset, with all indices falling within the recommended fit ranges (see Figure 2).

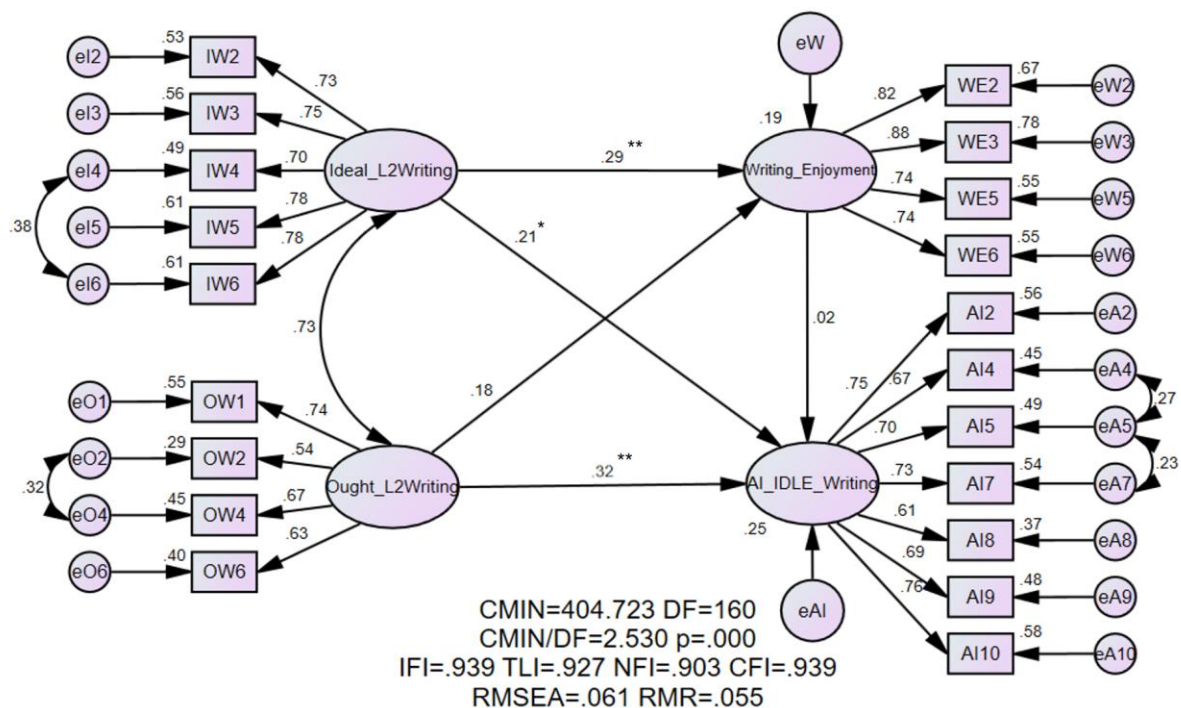
As shown in Table 4, three hypotheses were supported, while two hypotheses (i.e., Hypotheses 3 and 5) were rejected. Specifically, the ideal L2 writing self can make a positive and significant impact on L2 Writing Enjoyment (H1: $\beta = .29$, $p < .01$, $t\text{-value} = 3.08$) and can predict AI-IDLE-Writing positively and significantly (H2: $\beta = .21$, $p < .05$, $t\text{-value} = 2.18$). Additionally, the ought-to L2 writing self can play a positive and significant role in AI-IDLE-Writing (H4: $\beta = .32$, $p < .05$, $t\text{-value} = 3.11$). However, the ought-to L2 writing self cannot predict L2 Writing Enjoyment (H3: $\beta = .18$, $p = .063$, $t\text{-value} = 1.86$). It is worth noting that L2 Writing Enjoyment cannot serve as a predictor of AI-IDLE-Writing (H5: $\beta = .02$, $p = .780$, $t\text{-value} = 0.28$). R^2 observed in Figure 2 demonstrated that the ideal L2 writing self and the ought-to L2 writing self explained 19% of the variance in L2 Writing Enjoyment, while the three factors (i.e., the ideal L2 writing self, the ought-to L2 writing self, and L2 Writing Enjoyment) can make a joint effort to account for 25% of the variance in AI-IDLE-Writing. The R^2 values may indicate that the significant relationships in the structural model were in a moderate effect size range (Plonsky & Oswald, 2014).

Table 4
Hypothesis test results of the structural model.

Hypotheses	β	p	t-value (> 1.96)	Results
H1: Ideal_L2writing $\rightarrow$ Writing_Enjoyment	.29**	$p = .002$	3.08	Supported
H2: Ideal_L2writing $\rightarrow$ AI-IDLE-Writing	.21*	$p = .033$	2.18	Supported
H3: Ought_L2writing $\rightarrow$ Writing_Enjoyment	.18	$p = .063$	1.86	Unsupported
H4: Ought_L2writing $\rightarrow$ AI-IDLE-Writing	.32**	$p = .003$	3.11	Supported
H5: Writing_Enjoyment $\rightarrow$ AI-IDLE-Writing	.02	$p = .780$	0.28	Unsupported

Note: $p < 0.01^{**}$; $p < 0.05^*$

Figure 2.
The structural model of four factors



(R^2 : L2 Writing Enjoyment = 19%, AI-IDLE-Writing = 25%).

4.2. Qualitative findings

Analysis of interview data corroborated the quantitative findings, revealing that students with well-defined ideal and ought-to L2 writing selves utilised AI tools for various purposes, including AI-IDLE-Writing practices. Furthermore, while participants demonstrated appreciation for the English language and learning process, they reported low enjoyment in L2 English writing. Consequently, their engagement in AI-IDLE-Writing activities was primarily motivated by task completion rather than enjoyment. However, participants' writing motivation and enjoyment exhibited complex, non-linear interactions across different learning contexts. This dynamic process may also elucidate students' participation patterns in AI-IDLE-Writing activities.

4.2.1. How L2 writing motivation influences AI-IDLE-Writing

The qualitative data revealed that participants' ideal L2 writing self and the ought-to L2 writing self provided power for them to utilise AI technologies for engaging in AI-IDLE-Writing activities. Ten participants indicated that their motivation for English writing and the demands of real-world writing tasks encouraged or inspired them to probe the capabilities of new generative AI software (e.g., ChatGPT, Literary Words from the Heart [文心一言]) to meet their extracurricular English writing learning needs. For instance, Zhoujing¹, a first-year Materials Science and Engineering postgraduate who is passionate about publishing academic papers in Science Citation Index (SCI) journals, commented that:

¹ Pseudonyms were used for all interviewees.

AI is undoubtedly a groundbreaking technology, and its development holds great significance for human progress... It provides additional support for my writing, allowing me to engage in self-directed learning without needing to consult my teachers. My motivation for English writing comes from my desire to develop my English academic skills.

Zhoujing recognized the potential of AI as a “groundbreaking technology” that “holds great significance for human progress.” This positive view of AI and its role in supporting his writing journey indicated a personal interest in leveraging technology to enhance his skills, which aligns with an ideal L2 writing self. The use of AI provides “additional support” for his academic writing, enabling him to engage in “self-directed learning” without relying on others. This autonomous approach to skill development suggests an intrinsic motivation to improve, characteristic of an ideal L2 writing self. He also expressed a “desire to develop [his] academic skills.” This internal motivation to improve his writing abilities suggests the presence of an ideal L2 writing self, where the individual aspires to become a more proficient academic writer. On the contrary, Tangshan, a third-year postgraduate student majoring in Mechanical Engineering, who frequently used ChatGPT 4.0 for writing tasks related to research proposals for doctoral program applications, expressed that:

I am not an enthusiastic English writing learner, nor do I have a particular interest in English writing. I only use ChatGPT 4.0 when I need to complete a specific task. For example, when I was applying for a PhD program in the UK, I needed to submit a research proposal in English... I used ChatGPT 4.0 to understand the language expressions and formats used in research proposals, as well as to gain ideas or guidance for my own proposal.

In this particular case, Tangshan demonstrated an extrinsic instrumental motivation for English writing learning, driven by external pressures and specific task requirements (e.g., the goal of “successfully applying for a PhD program in the UK”) to achieve his academic advancement, rather than possessing an internalised ideal self as a proficient English writer. Similarly, other participants were primarily driven by their ought-to L2 writing selves, making them more likely to use AI technologies to complete tasks assigned in formal coursework rather than engaging in informal digital language writing learning. These participants were also motivated by the need to prepare for English proficiency certification exams through extracurricular writing practice. One such participant was Guanling, a fourth-year Finance undergraduate, who shared:

I believe that AI responds faster than the human brain and has a much larger knowledge base... I use the Tiangong AI tool for English writing learning primarily to prepare for English exams... Additionally, preparing for the CET-4 and CET-6 [College English Test-level 4 & 6] English certificates has influenced my motivation for English writing, as these certifications partly affect my graduation prospects by contributing to my overall credits.

Guanling’s interview excerpt portrayed a clear example of how an ought-to L2 writing self can shape a learner’s involvement with AI for self-directed writing practice beyond the classroom. Her externally driven motivation, rooted in exam preparation and certification requirements, has led her to use AI tools strategically to meet these demands. By engaging with AI beyond the classroom, she aims to enhance her writing performance and increase her chances of success in meeting these external demands. The AI tool serves as a

resource to help her navigate the language expressions, formats, and knowledge required for these specific writing tasks.

4.2.2. How L2 writing motivation interacts with enjoyment to shape AI-IDLE-Writing

The interviewees reported relatively low enjoyment in the process of English writing, generally being goal- or task-driven when engaging in AI-assisted informal writing learning activities beyond the classroom. For instance, Xincheng, a first-year postgraduate student majoring in Computer Science, remarked:

My interest in learning English writing stems from a curiosity about various foreign cultures... What I find intriguing is that the content of written texts carries cultural elements that aid in my understanding of the customs of English-speaking countries... My enjoyment lies in the desire to convey my ideas in English, which often takes the form of informal or even personal expressions, such as creative writing or songwriting in English.

Xincheng's description suggests that while he seeks to find enjoyment in English writing, it remains challenging for him. This transcript reflects an exemplary case of a learner motivated by the ideal L2 writing self. The learners' interest in cultural understanding and creative writing expression aligns with the social dimensions of the ideal L2 self (Dörnyei, 2009). In most cases, it is task-driven motivation that prompts him to use AI tools for extracurricular English writing practice. The qualitative data also revealed that L2 learners' enjoyment of English writing is intricately connected to their motivation. Some participants believed that sustained positive emotions (e.g., enjoyment) could, in turn, enhance their promotion-centred motivation for writing. For example, Cailing, a fourth-year undergraduate majoring in English education, commented:

I consider myself an active English writing learner... As a student, I am eager to enhance my English writing ability, driven by an intrinsic desire. As I am about to graduate and step into the role of an English teacher, my motivation is more focused on helping my students improve their English writing skills. I actively engage in learning writing because, only when I master it myself, can I teach them effectively... My interest in the writing process stems from receiving feedback after completing an essay.

The envisioning of Cailing's ideal L2 writing self - such as "helping my students improve their English writing skills I have learned" - can bring enjoyment (Cailing's a true sense of fulfillment), which in turn motivates learners to develop a stronger intrinsic drive for English writing ("I am eager to enhance my English writing ability and work hard" from Cailing). This motivates them to engage in extracurricular English writing practice using AI tools. The excerpt highlights proactive language learning behaviours, where the learner's motivation is shaped by their future teaching role and desire for self-mastery. These findings strongly resonate with Papi and Khajavy's (2021) conceptualisation of motivated learners maximising L2 opportunities and with Papi and Hiver's (2024) Proactive Language Learning Theory. In the interviews, we found that by aligning their ideal L2 writing self with their ought-to L2 writing self, participants were more likely to invest time and effort in AI-IDLE-Writing practices and, as a result, derive enjoyment from their writing.

5. Discussion

This study adopted an explanatory mixed-method approach to investigate the relationships among L2 writing motivational factors (i.e., ideal L2 writing self and ought-to L2 writing self), L2 writing enjoyment, and student involvement with AI-IDLE-Writing activities within the Chinese EFL context. Results of SEM demonstrated that students' ideal L2 writing self can positively predict both L2 writing enjoyment and AI-IDLE-Writing, while the ought-to L2 writing self can only influence AI-IDLE-Writing directly. Analysis of the interview data not only confirmed qualitative findings but also added nuances and contextual details to the quantitative results. It also disclosed a more dynamic and complicated mechanism of how L2 writing future selves, enjoyment and AI-IDLE-Writing intertwine with each other as learners negotiate their own learning contexts.

RQ1: To what extent, if any, does L2 writing motivation make an impact on Chinese EFL learners' participation in AI-IDLE-Writing?

The present study found that L2 writing motivation can significantly and largely influence students' involvement in AI-IDLE-Writing practices. Chinese EFL university students' ideal L2 writing self has a significant predictive effect on their participation in AI-assisted IDLE writing activities. This finding aligns with previous research on how students with clear ideal L2 selves are more likely to engage in IDLE practices in AI and non-AI digital learning environments (Lee & Lu, 2023; Liu, Darvin, & Ma, 2024a, 2024b; Zou et al., 2025a). This viewpoint is supported by the quantitative evidence (H2: the ideal L2 writing self $\rightarrow$ AI-IDLE-Writing, $\beta = .29$, $p < 0.01$, $t\text{-value} = 3.08$). Findings from the semi-structured interviews add further evidence. Participants clearly articulated future self-images, such as "appreciation for AI technologies", "desire for skills development", or "AI-mediated self-directed learning". These aspirations motivated them to actively use AI technologies to achieve their extracurricular English writing goals. On this ground, it could be indicated that the ideal L2 self motivates learners to engage in digital learning activities in the AI-mediated learning contexts (Liu, Darvin, & Ma, 2024a), in turn, utilising AI tools can significantly improve Chinese EFL learners' motivation, enjoyment and engagement in Yuan and Liu's (2025) quasi-experimental study.

Furthermore, this study disclosed that the ought-to L2 writing self can have a positive effect on learners' participation in AI-IDLE-Writing activities, contributing to the ongoing discussion on how the ought-to L2 self might encourage students to engage in informal language learning across contexts (Lee & Lu, 2023; Liu, Darvin, & Ma, 2024a, 2024b). Specifically, our observation was supported by quantitative data (i.e., H4: $\beta = .32$, $p < 0.05$, $t\text{-value} = 3.11$) and further corroborated by qualitative interview, demonstrating that the ought-to L2 writing self, as an external motivational variable, can directly prompt students to engage in AI-assisted writing tasks beyond the classroom. This engagement is often driven by pressures from standardised language exams, academic demands, time constraints, and the ease of use and interactivity of AI tools (Liu & Ma, 2024; Zhang & Liu, 2023). In the context of China's higher education system, the external pressure from CET 4 and CET 6 may strengthen students' ought-to L2 writing self, as they feel they "should" or "must" improve their English writing skills to successfully obtain these certificates. By comparing these findings with research on how AI can reshape language learning (Alghamdi, 2024; Barrot, 2023a, 2023b; Yang & Li, 2024), we further highlight that the heightened accessibility and functionality of AI-based language learning tools can amplify the influence of external motivational factors like the ought-to L2 writing self.

The study advances the theoretical understanding of L2 motivation by demonstrating that the ideal L2 writing self operates as an intrinsic driver, enabling learners to leverage generative AI tools for agentic, creative writing tasks beyond formal instruction—a phenomenon suggesting AI's capacity to amplify identity-driven agency in language learning ecosystems. Simultaneously, it extends the ought-to L2 self construct by revealing how high-stakes testing regimes and AI tool affordances co-constitute extrinsic motivational pathways, positioning AI-IDLE environments as contested spaces where institutional pressures and technological mediation dynamically reconfigure learners' goal-oriented behaviors.

RQ2: To what extent, if any, does L2 writing enjoyment have an effect on their involvement in AI-IDLE-Writing?

This study reveals that L2 writing enjoyment cannot directly predict students' practises of AI-IDLE-Writing, which contrasts with the findings of Liu, Darwin, and Ma (2024a) in their research on AI-IDLE. The research result diverges from Liu and Fan's (2024) observation that reveals Chinese university students with a greater sense of enjoyment are more willing to communicate with others in AI-mediated speaking environments, as well as from Hang and Zhang's (2025) investigation, which demonstrates that university students who enjoy L2 Chinese speaking more are more likely to enhance their willingness to communicate in online digital learning settings. This discrepancy may be attributed to the distinct nature of writing as a complex language output activity that requires a comprehensive and integrated knowledge base (Guo & Wang, 2023; Naghdipour, 2022; Shin & Lee, 2024; Solhi et al., 2024; Teng & Zhang, 2018; Wilson et al., 2024; Zhu et al., 2022). The emotional experience and learning approaches associated with writing may differ from those in the general language learning process (Li et al., 2023; Solhi et al., 2024; Zhu et al., 2022). In the current study, we adopted a focused and in-depth L2 writing perspective on AI-IDLE to examine the predictive effect of enjoyment on AI-mediated informal L2 writing practices. While L2 writing enjoyment may not be the primary driver of learners' participation in AI-IDLE-Writing activities, other factors, such as the ought-to L2 writing self, appear to play a more significant role. This exploration extends beyond the single dimension of enjoyment, offering a comprehensive understanding of L2 writing learning motivation by highlighting the significant roles of internal self-improvement and external academic pressure in motivating participation in extracurricular L2 writing practice (Shafiee Rad et al., 2024; Tahmouresi, & Papi, 2021; Teng, & Zhang, 2018). As such, we argue that by providing a nuanced perspective on the role of emotional factors in AI-assisted informal L2 writing learning, it is essential to further explore the complex interplay between learner emotions, motivational selves, and AI-mediated self-directed language learning behaviours.

This study challenges the universal applicability of positive emotion frameworks in AI-mediated language learning by proposing domain-specific motivational mechanisms for L2 writing, where the cognitive complexity of composition processes disrupts direct enjoyment-practice correlations—necessitating theoretical models that distinguish between general language acquisition and skill-specific affective architectures. It theoretically reconceptualises AI-IDLE ecosystems that prioritise dynamic self-concept negotiations (ideal/ought-to L2 writing selves) over emotional valence, positioning AI tools as amplifiers of identity-driven agency that mediate between learners' internal aspirations, external academic regimes, and the epistemic demands of advanced literacy development.

6. Concluding remarks, limitations and implications

This study examined the role of L2 writing motivation (i.e., the ideal L2 writing self and the ought-to L2 writing self) and L2 writing enjoyment in Chinese EFL university students' involvement with AI-IDLE-Writing practices. We believe this study provides valuable insights into the mechanisms by which motivational and emotional factors influence students' utilisation of AI technologies for informal writing learning activities.

There are some limitations that warrant attention. Firstly, while this study primarily focused on the L2 writing context, language learning encompasses other skills (e.g., reading, listening, speaking, translation, and interpreting). These different language skills may lead learners to adopt varied learning behaviours, resulting in different mechanisms when using AI-assisted language learning. Secondly, this study employed a cross-sectional approach, which has a limitation in depicting patterns of the dynamic changes in learners' L2 writing motivation and writing enjoyment over time in the context of using AI technologies for AI-IDLE-Writing practices. Thirdly, the study focused solely on motivational and emotional factors influencing students' participation in AI-IDLE-Writing activities. However, students' engagement in AI-IDLE-Writing activities may also be related to other individual differences (e.g., self-efficacy, beliefs, L2 grit) or contextual factors (e.g., teacher support, teacher enthusiasm, learning atmosphere). Additionally, the cross-sectional nature of the study limits causal inferences, and self-reported data may introduce bias, thereby recommending the need for experimental studies to explore the mechanism of causal inferences.

The findings propose several implications for future teaching practices. Recognizing that the ideal L2 writing self serves as a significant and powerful motivational factor in extracurricular English writing learning in the age of AI, teachers should foster a classroom environment that encourages creativity and embraces diversity in writing. With regard to strike a balance between creativity and the use of AI technologies, we propose several strategies that teachers can adopt to cultivate a classroom environment that nurtures creativity while harnessing the benefits of AI. For instance, teachers can design writing tasks that necessitate students to employ higher-order thinking skills (Wu et al., 2024), such as evaluation and synthesis, which are less likely to be automated by AI. Second, teachers may incorporate collaborative writing activities that underscore the social and interactive aspects of writing, which are more difficult for AI to replicate (Dobao, 2012). By facilitating peer feedback and group brainstorming sessions, educators can foster a sense of community and encourage students to exchange diverse ideas, thereby promoting their creativity. Third, teachers can provide explicit guidance on using AI-generated text as a starting point for students' writing, encouraging them to critically adapt the AI output to fit their unique voices and viewpoints. Additionally, students should be encouraged to express their unique perspectives. Teachers can motivate students to participate in extracurricular English writing clubs or guide them in using AI technologies for cross-cultural writing projects. Similarly, as the ought-to L2 writing self has a positive impact on students' participation in AI-IDLE-Writing practices, teachers can reinforce this by sharing the writing experiences and achievements of exemplary students, thereby providing positive role models. This can help students recognize that they "should" make efforts to enhance their writing skills, strengthening their ought-to L2 writing self. Encouraging student engagement in AI-IDLE-Writing not only necessitates an advocating and motivating writing environment but also the active back of teachers, along with peer and role model encouragement. This collective support can help students realise the value of AI-assisted writing in achieving their academic, personal, career, and professional development goals.

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